Problem 2

Goal: $X = \left\{ \text{Primes with remainder 3 when divided by 4} \right\}$

This set is infinite

Strategy: Given any finite subset

$\{p_1, p_2, \ldots, p_r\} \subseteq X$

find $q \in X$ s.t. $q \notin \{p_1, p_2, \ldots, p_r\}$

Consider: $N = 4 p_1 p_2 \cdots p_r - 1$

Then: (a) $N = \underbrace{q_1 q_2 \ldots q_m}_{\text{prime factors}}$

(b) N has a remainder of 3 when divided by 4

Modular arithmetic

Def$^{\underline{n}}$ $n \in \mathbb{Z}$

$$\mathbb{Z}/n\mathbb{Z} = \{0, 1, \ldots, n-2, n-1\}$$

"zee-mod-n-zee"

with arithmetic given by:

$a \in \mathbb{Z}/n\mathbb{Z}$, $b \in \mathbb{Z}/n\mathbb{Z}$

$a+b$: Take the integer $a+b$ ($a \cdot b$,
$a \cdot b$ $a-b$) & then
$a-b$ look at the remainder
when divided by n

If $m \in \mathbb{Z}$ and we want to think of it as an element of $\mathbb{Z}/n\mathbb{Z}$ then we will write $m \pmod n$

Example: ① $n = 4$

$\mathbb{Z}/4\mathbb{Z} = \{0, 1, 2, 3\}$

$9 \pmod 4 = 1 \pmod 4$

or

9 is congruent to 1 mod 4 $9 \equiv 1 \pmod 4$

congruent

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

·	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

Example 2

$n = 5$

·	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

$$3 = 2^{-1}$$

$$2 = 3^{-1}$$

Fields

Def^n

A field is a set F equipped with

(Addition)

$$+ : F \times F \longrightarrow F$$
$$(a, b) \longmapsto a+b$$

(Multiplication)

$$\cdot : F \times F \longrightarrow F$$
$$(a, b) \longmapsto a \cdot b$$

(Zero)

$0 \in F$

(One)

$1 \in F$

Satisfying:

Addition	Multiplication	Together
• $a+b=b+a$ • $a+0=0+a$ $=a$ • $\forall a, \exists -a$ s.t. $a+(-a)=0$ • $(a+b)+c$ $=a+(b+c)$	$a \cdot b = b \cdot a$ $a \cdot 1 = 1 \cdot a = a$ • $\forall a \neq 0, \exists a^{-1}$ s.t. $a \cdot a^{-1} = a^{-1} \cdot a = 1$ • $a \cdot (b \cdot c)$ $= (a \cdot b) \cdot c$	$a \cdot (b+c)$ $=$ $a \cdot b + a \cdot c$

Notation

(a) Cartesian product

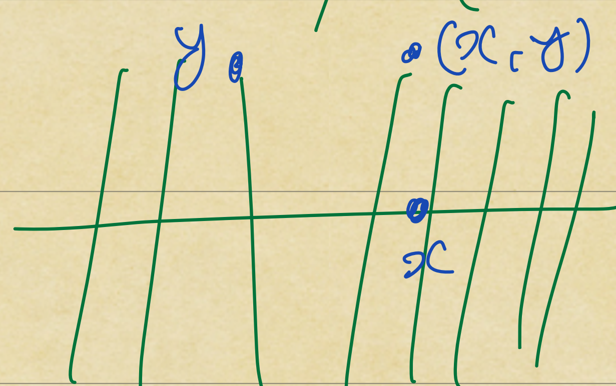
X, Y : two sets

$$X \times Y = \{ \underbrace{(x, y)}_{\text{"ordered pair"}} : x \in X, y \in Y \}$$

Giving two "co-ordinates"
the first is in X
& the second is in Y

<u>Example:</u> $\mathbb{R} = X$, $\mathbb{R} = Y$

$\mathbb{R} \times \mathbb{R}$:



(h) Functions

X, Y : two sets

A function from X to Y

$f : X \longrightarrow Y$

is a rule assigning to every element $x \in X$ an element $f(x) \in Y$

We write $f : x \underbrace{\longmapsto}_{\text{"maps to"}} f(x)$

X : domain of f

Y : codomain of f